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NPS/003/027 – Technical Specification for 24V Telecontrol Battery and Charger Systems

1. Purpose

This document is the technical specification for 24V battery and charger systems for use by CE Electric UK.

This document supersedes the following documents, all copies of which should be destroyed.

Ref	Version	Title
DSS/007/019	1	Specification for 24V Telecontrol Battery and Charger

2. Scope

This specification details the requirements for 24V battery and charger systems for use in CE Electric UK substations where DC supplies are required for the remote operation and monitoring of plant and equipment via SCADA.

3. Technical Requirements

Battery and charger systems shall be supplied as specified in Appendix 1. Battery and Chargers shall be entirely maintenance free.

The technical specification of any system must be declared using the table in Appendix 2.

The equipment shall comply with the current editions of BS EN 6290-4 and ENATS 50-18 except where varied by this specification and associated appendices.

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4. References

4.1 External Documentation

Reference	Title
BS EN 6290-4	Lead-acid stationary cells and batteries. Specification for classifying valve regulated types.
ENATS 50-18	Design and application of ancillary equipment

4.2 Internal documentation

Reference	Title
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5. Definitions

Term	Definition
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6. Authority for issue

6.1 Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and authorisation.

		Sign	Date
V Gorny	Standards Engineer	V Gorny	05/10/2009

6.2 Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
M Nicholson	System Strategy	M Nicholson	21/10/2009

6.3 Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
K Midgley	Standards Engineer	K Midgley	26/10/2009

6.4 DBD Assurance

I sign to confirm that this document has been assured for issue on to the DBD system

		Sign	Date
S.Johnson	DBD Administrator	S Johnson	02/11/2009

6.5 Approval

Approval is given for the content of this document

		Sign	Date
C.Holdsworth	Standards and Assessment Manager	C.Holdsworth	27/10/2009

6.6 Authorisation

Authorisation is granted for publication of this document

		Sign	Date
C.Morgan	Head of Procurement	C.Morgan	27/10/2009

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Appendix 1: Technical Specification for 24V battery and battery-charger system.

Battery Output Voltages and Load Duty	
Nominal voltage	24V
Float voltage	27.36V
Minimum open circuit voltage (OCV) after specified discharge profile	23.6V
Maximum voltage under all conditions	28V
Battery standing load	5A
Battery discharge profile	5A / 6 hr
Other Battery Design Factors	
Standard	
Type	
Service life	10 years
Number of cells	
Battery Capacity	To meet the discharge profile indicated
Ageing factor	Typically 1.25 lead acid
Minimum topping-up period	N/A
Rated temperature	15°C
Temperature range	-5°C, + 40°C
Mounting	Integrated Battery and Charger Wall Mounted Cubicle
Connectors	Shrouded
Isolation facilities required	From charger and distribution board
Battery earthing	The battery shall be unearthed
Low Voltage Disconnect	A load disconnect feature is required to prevent battery damage.
Charger Supply	
AC Supply	230v AC single phase
Supply Isolation	Double Pole Switch and Fuse or MCB
Supply Range	+10% / -6%
Nominal supply frequency	50Hz
Supply frequency range	47 / 52 Hz
Charger	
Type	Constant voltage, single stage, temperature compensated.
Ambient temperature range	-5°C, + 40°C
Float Voltage	27.36V
Maximum voltage under all conditions	28V
Voltage adjustment facility	Float voltage should be adjustable however such controls should not be on the front facia.

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Rated charging current	To cover standing load as specified plus adequate charging current
Charger voltage ripple / transient output	≤2mV (RMS) CCITT weighted No voltage transient shall occur on restoration of charger mains supply.
Battery/Charger Output Variation for :- Input voltage of 230v +6% / -10% Frequency of 50Hz +/- 10% Output load between 0 – 100%	27.36V +/- 0% (@ output terminals)
Transformer screening	Earth metal screen between primary and secondary windings.
Secondary wiring	ENATS 50-18 Not less than 2.5 sq mm 7/0.67mm for battery circuits and 1.5 sq mm 30/0.25 for other circuits.
System Monitoring	
Supply healthy / fail alarm	Local indication plus spare contact
Charger healthy / fail alarm	Local indication plus spare contact
Battery low voltage disconnect	Local indication plus spare contact
Monitoring alarm protection	2A HRC fuse
Remote voltage indication	Battery output connected to terminals via 1.82kΩ precision resistor (+ve) for voltage monitoring
Instruments	
Battery output voltmeter	Local indication
Charger output ammeter	Local indication
Cubicles and Battery Stands	
General arrangement	Integrated Battery and Charger Wall Mounted Cubicle
Design life	40 years
Cubicle cable entry	Bottom
Cubicle Colour	Light grey
Exposed conductors	Shrouded
Doors	Lockable
Distribution Board	
Number of ways	Min 2
Rating of ways	16A
Labels	
Fuses	
Type	Cartridge type: BS88

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Appendix 2 - SELF CERTIFICATION CONFORMANCE DECLARATION

24V Battery and Charger Systems required to be supplied against this specification shall comply with the latest issues of the relevant ENATS, British and International Standards specified. The following tables are intended to amplify and/or clarify the requirements of elements of these Standards but do not preclude meeting all requirements of the standards. .

The manufacturer shall declare conformance or otherwise, clause by clause, using the following levels of conformance declaration codes.

Conformance declaration codes

N/A = Clause is not applicable/ appropriate to the product

Cs1 = The product conforms fully with the requirements of this clause

Cs2 = The product conforms partially with the requirements of this clause

Cs3 = The product does not conform to the requirements of this clause

Cs4 = The product does not currently conform to the requirements of this clause, but the manufacturer proposes to modify and test the product in order to conform.

Instructions for completion

- When Cs1 code is entered no remark is necessary.
- When any other code is entered the reason for non-conformance shall be entered.
- Prefix each remark with the relevant 'BS EN' 'IEC' or 'ENATS' as appropriate.

Manufacturer:

Product Reference:

Details of the Cable Type (Voltage, Conductor Type and Size)

Name:

Signature:

Date:

NOTE:

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TECHNICAL SPECIFICATION FOR 24V BATTERY AND CHARGER SYSTEMS				
		Clause / Requirements	Conformance Code	Remarks / Comments
System Output Characteristics				
	Nominal voltage	24V		
	Float voltage (max acceptable)	27.36V		
	Minimum open circuit voltage (OCV) after specified discharge profile	23.6V		
	Maximum charger voltage under all conditions	28V		
	Maximum open circuit voltage	State		
Battery				
	Standard	State (BS/IEC?)		
	Type of cells	State		
	Service life	10Years		
	Aging factor applied	1.25 lead acid		
	Minimum topping-up period	N/A		
	Number of cells in battery	State		
	Float voltage per cell	State (V)		
	Rated temperature	15 °C		
	Temperature range	-5 / +40 °C		
	Cell mounting:	State		
	Connectors	Shrouded		
	Isolatable from both dist board and charger	Required		
	Battery Earthing	Unearthed		
	Remote Voltage Monitoring Facility	Required		
	Auto Battery Disconnect	State		

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TECHNICAL SPECIFICATION FOR 24V BATTERY AND CHARGER SYSTEMS				
		Clause / Requirements	Conformance Code	Remarks / Comments
	Charger Supply			
	AC Supply	230V / 1 Phase		
	Supply Isolation	DP Switch and Fuse or MCB		
	Nominal supply frequency	50Hz		
	Supply frequency range	47-52Hz		
	Charger			
	Type	Constant Voltage		
	Ambient temperature range	-5 / +40 °C		
	Float Voltage	27.36 V		
	Maximum Voltage	28V		
	Voltage adjustment facility (float - located behind facia).	Required		
	Battery/Charger Output Variation for :- Input voltage of 230v +6% / -10% Frequency of 50Hz +/- 10% Output load between 0 – 100%	27.36V +/- 0% (@ output terminals)		
	Transformer screened as per specification	Required		
	Secondary Wiring	ENATS 50-18		

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TECHNICAL SPECIFICATION FOR 24V BATTERY AND CHARGER SYSTEMS			
	Clause / Requirements	Conformance Code	Remarks / Comments
System Monitoring			
Mains input healthy / fail	Local+Contact *		
Charger output healthy / fail	Local+Contact *		
Alarm circuit fuse rating	2 A		
Battery Damage (Auto Disconnect / Restore)	Auto Disconnect		
Instruments			
Battery output voltmeter	Required		
Charger output ammeter	Required		
Cubicles and Battery Stands			
Design life	40 Years		
Mounting Arrangement	Wall		
Cubicle cable entry position	Bottom		
Cubicle Colour	Light Grey		
Exposed Conductors	Shrouded		
Doors I	Lockable		
Dimensions (state H x W x D in mm)			
Distribution Board			
Number of ways	Minimum 2		
Rating of ways	16 A		
Fuses			
Type	BS88		

* - Local indication + “spare” contact available for remote alarm